

Search Engines 2026

Grading and Project

Abstract

Your final grade for this course is an assessment of your project report & your performance at the oral exam. This document describes how your final grade is computed (Section 1) and the project specifications (Sections 2-9). Important dates are summarised in Table 1.

1 Your final grade

Your final grade for this course is an assessment of the final report that you submit for this project (on 4 June 2026) & of your performance at the oral exam (to be held on week 25). Both project report & oral exam are compulsory. This means that if you submit the project report, but do not show up for the oral exam, you will be graded as *absent*. Similarly, if you do not submit the project report, you are not eligible to take the oral exam, and you will be graded as *absent*.

1.1 Project

The project¹ is composed of six parts. Each part corresponds to the material taught in the course lectures. All parts of this project are compulsory, unless explicitly marked as **[OPTIONAL]** in blue font. The project will be graded as a whole. The project should be completed individually (no group submissions).

Halfway through the course (week 21), you have the option to submit your midterm project report for feedback and grading (see Section 5). If you do this, then you will receive a grade and feedback on your grade (what was good in your report, and what you can improve in order to increase the grade). All this (midterm report submission, grading and feedback) is optional and aims solely to help you understand how you can improve your work and get a better grade. The midterm report grade is **not** part of your final grade for this course.

¹Small modifications in the project description may be made throughout the course.

WHEN	WHAT	
Week 21, Wed 20 May by 10:00	Unseen queries released (Sec. 7)	
Week 21, Wed 20 May by 16:00	Submit midterm report (Sec. 2-5)	Optional
Week 22, Wed 27 May by 16:00	Feedback & grades on midterm report released	
Week 22, Wed 27 May by 16:00	Submit 3-5 runs on unseen queries (Sec. 7)	Compulsory
Week 22, Thu 28 May by 17:00	Unseen queries results released (Sec. 7)	
Week 23, Thu 4 June by 16:00	Submit final report (Sec. 9)	Compulsory
Week 25	Oral exam	Compulsory

Table 1: Important dates.

1.2 Report submission details

1.2.1 Midterm report submission [OPTIONAL]

For the midterm report, you must submit the first part of the project report (as described in Sec. 5). You should only submit the report. You should not submit code or any other files:

- **WHEN YOU SUBMIT:** Week 21, Wednesday 20 May by 16:00
- **WHERE YOU SUBMIT:** Absalon
- **WHAT YOU SUBMIT:** PDF, using the ACL template², no longer than 4 pages, excluding references.

Please state your name clearly at the top of the page (the submission should not be anonymous).

1.2.2 Final report submission

For the final project submission, you must submit both your final report of the whole project (see Sec. 9) and your code:

- **WHEN YOU SUBMIT:** Week 23, Thursday 4 June by 16:00
- **WHERE YOU SUBMIT:** Digital Exam
- **WHAT YOU SUBMIT:** (1) PDF, using the ACL template, no longer than 9 pages, excluding references; (2) .zip file containing the code of your experiments and documentation (readme file) on how to run it.

For both midterm and final report, it is compulsory that you use the ACL template. Please state your name clearly at the top of the page (the submission should not be anonymous).

1.3 Oral exam

The oral exam consists of a 10-minute presentation (by you) of your project to the examiners, followed by questions by the examiners. The examiner questions

²<https://2021.aclweb.org/downloads/acl-ijcnlp2021-templates.zip>

can cover your project, but also material covered in the lectures and not included in this project.

You need to have prepared slides for your 10-minute presentation, and make sure that you can connect your machine to the projector for the presentation.

2 Dataset & Indexing (week 17)

The overall goal of this project is to put together a search engine, train and evaluate it. In this section, we focus on the dataset and the index of the search engine.

2.1 Dataset

We will use the Natural Questions dataset³ for document ranking. Natural Questions (NQ) is an automated question-answering and retrieval benchmark based on real user queries issued to the Google search engine, paired with annotated answers from Wikipedia articles. It is widely used to evaluate open-domain retrieval given natural language questions. For this project, we will provide you with a sample of ~150,000 documents. Each document comes with text that contains both a title and body. You will be given a sample of 10,000 queries and their corresponding relevance assessments for training purposes. Later in the project (Section 7), you will be provided with 5,000 new, unseen queries, which you will use to test your trained search engine. Relevance assessments are on a 3-point scale:

- **Random non-relevant (0)** - This document does not contain useful information for the query
- **Partially relevant (1)** - This document contains some relevant information but is incomplete, marginal, or not focused on the query
- **Relevant (2)** - This document fully addresses the given query

On Absalon, you can find the documents⁴, training queries⁵ and their relevance assessments (qrels)⁶. You will not have access to the relevance assessments of the unseen test queries.

To include in your report:

1. Present (in a table and/or figure) the following analysis of the dataset:

- The total number of documents and queries in the version of the dataset you have been given.
- The min, max, median and average length of the queries (in terms of number of words or characters).
- The min, max, median and average length of the documents (in terms of number of words or characters).

³<https://github.com/google-research-datasets/natural-questions>

⁴https://absalon.ku.dk/courses/89586/files/10697122?module_item_id=3040440

⁵https://absalon.ku.dk/courses/89586/files/10697121?module_item_id=3033373

⁶https://absalon.ku.dk/courses/89586/files/10697151?module_item_id=3040442

- The distribution of document lengths (e.g. histogram or box plot).
 - The distribution of query lengths (e.g. histogram or box plot).
2. Briefly discuss these statistics. Do you see anything unusual?

You should also download and familiarise yourself with one of the following IR system libraries: Terrier⁷, Indri⁸, Elasticsearch⁹, Anserini¹⁰, etc. We strongly recommend using PyTerrier¹¹, a Python framework that acts as a wrapper and interface for the Java-based Terrier Information Retrieval platform. Students have reported several issues with other frameworks in previous iterations of this course. Additionally, you can use the free version of Google Colab¹² for running the experiments.

2.2 Indexing

The objective of this section is to help you build the index of your search engine. The following tasks should be completed:

- Load and index the dataset. We suggest that you use existing tools and libraries (e.g., Terrier, Indri, Elasticsearch, Anserini, etc.) to build your index. Python wrappers for different libraries are also available (e.g., PyTerrier¹³, Pyserini¹⁴, Elasticsearch-py¹⁵, etc.).
- You should build four different versions of your index: (1) full index, (2) stopwords removed, (3) stemming, (4) stopwords removed + stemming.

To include in your report:

1. After building each version of your index, report in a table, for each index version, the number of documents indexed, number of unique terms, total number of tokens^a, index storage size, how much time it took to build each version of your index, and how much time (average searching time) it takes to process a query with each version of your index.
2. Include in the report all the important details regarding the implementation of your index, including any preprocessing of the data.
3. Briefly discuss: What are the advantages and disadvantages of each indexing approach? What are the most and least efficient (indexing time or retrieval time) indices?

^a**Tokens:** Raw split text units (typically by whitespace delimiters). e.g. Given the document "banana banana", we have the tokens ["banana", "banana"], with a token count of 2. **Terms:** Unique tokens. Given the same example as above, we have a term count of 1 (["banana"]).

⁷<http://terrier.org/>

⁸<http://www.lemurproject.org/indri.php>

⁹<https://www.elastic.co/>

¹⁰<https://github.com/castorini/anserini>

¹¹<https://pyterrier.readthedocs.io/en/latest/>

¹²<https://colab.research.google.com/>

¹³<https://pyterrier.readthedocs.io/en/latest/>

¹⁴<https://github.com/castorini/pyserini>

¹⁵<https://github.com/elastic/elasticsearch-py>

3 Ranking Models and Evaluation (week 18, 19)

Once you have the index, you can start implementing your ranking models. You should do the following: Tune and run BM25 and a Language Model (LM). Given the time requirements, we require that you do this **only for the two indices with the stopwords removed**¹⁶, separately. You are allowed to use existing search libraries. The same library that you have used to build your index will likely have a version of BM25 and LM.

To tune your ranking models, you should use all your training queries. After finding the best configuration for your model, you should use that model (tuned model) on the unseen queries that we will release on week 21.

You should tune with respect to one evaluation measure only (either MRR or NDCG@10). We require that you use `trec-eval`¹⁷ for evaluation.

To include in your report:

1. Present in a table, for each ranking model, and for each version of the index **with the stopwords removed**, NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs, and mean response time (the time of getting the evaluation score of a query). If you split the training queries (for example into train and validation sets), present the results of the above metrics on each of the sets.
2. Include in the report all the important details regarding the implementation of your models, for example the library you used, how you tuned the parameters of your model(s), etc.
3. Briefly discuss and compare different retrieval approaches and the limitations of different evaluation measures. What limitations do you identify in different evaluation measures? Which combination of index and ranking model is the most effective? Which one is the least effective? What affects effectiveness more, the index or the ranking model?

Different models will perform differently also depending on the evaluation measure. It is up to you to decide whether to report the results in tables, plots, box-plots, etc. Remember to give insights on which models and evaluation measures work the best and why, highlighting your design decisions.

[OPTIONAL] You are not limited to the above ranking models and evaluation measures; you can use further ranking models and evaluation measures and explore different cut-offs. Moreover, you can tune and evaluate also indices without the stopwords removed.

4 Relevance feedback and Query Expansion (week 20)

The goal of this week is to add a pseudo relevance component to your search engine.

You should use pseudo relevance feedback to expand the query and run it with BM25 and LM. You should tune some of the parameters of pseudo relevance

¹⁶(i) index with stopwords removed, and (ii) index with stopwords removed + stemming.

¹⁷https://github.com/usnistgov/trec_eval

feedback using an evaluation measure of your choice, e.g., if you use RM3 pseudo relevance feedback you can tune only *fb_terms* and *fb_docs*. You should tune with respect to one evaluation measure only (either MRR or NDCG@10). You should use your single best performing model-index combination as the backbone for the retrieval of the top-K documents to expand the queries and for the final retrieval.

In addition, you should use word embeddings to do query expansion as done by Kuzi et al.¹⁸[5]. A popular word embedding choice is the word2vec pre-trained on Google News corpus,¹⁹ but you can decide on another embedding. Another choice of embeddings is using contextualized word embeddings, such as BERT [3].

To include in your report:

1. Report in a table the results of the above experiments with respect to NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs. Also report your methods' mean response time (the time of getting the evaluation score of a query).
2. Plot (a) the number of expansion terms used for pseudo relevance feedback versus an evaluation measure of your choice (while keeping the number of expansion documents fixed, e.g. to 10), and (b) the number of expansion documents used for pseudo relevance feedback versus an evaluation measure of your choice (while keeping the number of expansion terms fixed, e.g. to 5). Then discuss how varying those two parameters (number of expansion terms and number of expansion documents) affects retrieval performance.

5 Midterm submission (week 21) [OPTIONAL]

You can submit a report including all the components of the project that have been requested up to this point. You can submit the midterm report no later than **Wednesday, May 20 2026 at 16:00**. The format of the report should be a PDF document using the ACL template, no more than **4 pages** (not including references).

6 Query expansion with LLMs (week 21)

The aim of this week is to perform query expansion by prompting a Large Language Model (LLM) [7, 4]. You must use any one of these three prompting strategies: i) zero-shot, ii) few-shot, and iii) chain-of-thought (COT) prompting. You can use any free, publicly available LLM like llmware/bling-1b-0.1 or google/flan-t5-(small/large) (both available through the Python library `huggingface`²⁰). Running LLMs for inference can be time consuming so the usage of a GPU, either locally or through free services like Colab²¹ or Kaggle notebooks²², is advised (bling-1b-0.1 prompting should run approximately under 1 second per

¹⁸<https://dl.acm.org/doi/pdf/10.1145/2983323.2983876>

¹⁹<https://github.com/mmihaltz/word2vec-GoogleNews-vectors>

²⁰<https://huggingface.co/>

²¹<https://colab.google/>

²²<https://www.kaggle.com/code>

prompt, whereas flan-t5 under 0.25 second per prompt, both tested on Kaggle GPUs).

To include in your report:

1. State which LLM you used. You should state not only its name, but also its version, size (in terms of parameters but also size of data it was trained on, if this is available), and include the URL where it can be accessed.
2. Report in a table the results of these experiments with respect to NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs, and the prompting time per query. The table should also include baselines without query expansion, so that we can compare them.
3. Discuss briefly: Which query expansion approach (pseudo relevance feedback, word embeddings, LLMs) works best and why? Are there some queries for which neither of the query expansion methods improves retrieval over the baselines?
4. [OPTIONAL] Sort the NDCG@10 score of each query for your single best performing LLM-QE run and plot the resulting score distribution in a histogram. Do the same for your single best baseline. Discuss the resulting histograms.

7 Evaluation on unseen queries (weeks 21, 22)

The goal of this section is to evaluate your search engine on unseen queries. We will release 5,000 unseen queries on Absalon by Wednesday May 20, 10h00. You should choose your 3-5 best configurations of (a) index, (b) ranking model, (c) relevance feedback or not, and run this configuration on the unseen queries. You will not be able to evaluate your search engine on these unseen queries, because we will not release the corresponding relevance assessments. Instead, you must submit your runs on the unseen queries to us on Absalon, and we will evaluate them and then release the corresponding scores. Details of this are described below.

7.1 What you must submit

You must submit no less than 3 runs and no more than 5 runs on the unseen queries. Your run files must be in trec-eval format, each of them in a separate .txt file, and all of them uploaded as one .zip file on Absalon by **Wednesday May 27 at 16h00** at the latest. You must choose a name for each run. The name should be the filename of the .txt file that contains that run. Each run should have a different name. Run names should be in the following format: {any 3 small letters of your choice}{any three numbers between 0-9 of your choice}{any three capital letters of your choice}{any three numbers between 0-9 of your choice}. For example, the following are valid run names:

- dog357IJS302
- lod547ASE667
- qok312POW987
- aqw769GHH095

The following are examples of invalid run names:

- dg357IJS302
- lod547.ASE
- 312POW987qok
- aqw7650GHH095

We will evaluate all the runs we receive and we will release the results on Absalon by Thursday 28 May at 17h00 at the latest. You should make sure that you remember the name of your own runs, so that you can look at their performance and **include a discussion of this in your report**.

7.2 Format of your runs

The submission format of your runs must follow the standard TREC run format. The submission format of a ranked result list (run) is as follows:

```
qid Q0 docno rank score tag
```

The fields should be separated with a white space. The width of the columns in the format is not important, but it is important to include all columns and have some amount of white space between the columns. The above fields are:

- **qid**: the query number;
- **Q0**: unused and should always be Q0;
- **docno**: the official document id number returned by your system for the query **qid**.
- **rank**: the rank where the document is retrieved;
- **score**: the score (integer or floating point) that generated the ranking. The score must be in descending (non-increasing) order. The score is important to handle tied scores (**trec_eval** sorts documents by their scores values and not their ranks values);
- **tag**: your run name, in the format described above. **Each run should have a different tag.**

An example of a run is shown below:

```
1 Q0 doc1 1 14.8928003311 dgp357IJS302
1 Q0 doc2 2 14.7590999603 dgp357IJS302
1 Q0 doc3 3 14.5707998276 dgp357IJS302
1 Q0 doc4 4 14.5642995834 dgp357IJS302
1 Q0 doc5 5 14.3723001481 dgp357IJS302
...
```

The submitted runs should contain at most the top 1,000 documents for each query, with all the queries in each file. The submission file must be in **.zip**.

7.3 Evaluation of Runs

We will evaluate the submitted runs with the following measures: NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs. We will post the results of the evaluation on Absalon by Thursday 28 May at 17h00. If a run is not in the correct format and cannot be processed by trec-eval, we will ignore it.

To include in your report:

1. Present in a table, for each run on unseen queries, the NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs (use the exact values of these scores as released by us).
2. Include in the report all the important details regarding the implementation of each of your runs on the unseen queries, so that someone reading your report can manage to replicate precisely your runs.
3. Briefly discuss and compare the performance of your runs on the unseen queries. What limitations do you identify in different evaluation measures? Which combination of index and ranking model is the most effective? Which one is the least effective? What affects effectiveness more, the index or the ranking model?
4. Find the 5 worst performing seen queries (according to MRR), and present their text and their respective MRR in a table. Then reflect on why you think these 5 queries have the worst performance. Do the same for the 5 best performing seen queries.
5. Reflect on the differences in effectiveness between the seen and unseen queries.

Different models will perform differently also depending on the evaluation measure. It is up to you to decide whether to report the results in tables, plots, box-plots, etc. Remember to give insights on which models and evaluation measures work the best and why, highlighting your design decisions.

8 Refining Document Representation and Re-ranking (week 23)

You should use the best (tuned) version of BM25 and LM from the mid-term report to generate an initial ranking (separately for BM25 and for LM), and then re-rank the top K^{23} documents using contextualized embedding approaches such as BERT [2], or Sentence-BERT [6]. To do so, you have to build the query and document representations using these approaches and then compute the similarity score between them. The similarity score between each document and query can be used to re-rank the documents. You can also get some inspiration from [1].

To include in your report:

1. Present in a table, for each model, separately before and after reranking, the NDCG, MRR, Precision and Recall at 5, 10 and 20 cutoffs.
2. Include in the report all the important details regarding the implemen-

²³Typically, $K \in \{100, 500, 1000\}$.

tation of each of your reranking runs, so that someone reading your report can manage to replicate precisely your runs.

3. Briefly discuss and compare the performance of your reranking runs. Does re-ranking through embedding approaches lead to better retrieval? What limitations do you identify?

4. Find the 5 queries that benefit most from reranking, and the 5 queries that are hurt the most from reranking. Present their text and their respective MRR in a table. Then reflect on why you think these queries have benefited or suffered the most from reranking.

9 Your final report

Your final report should include everything that appears framed in boxes and marked: **To include in your report** in Sections 2 - 8.

Helpful tips for the discussion: Try to describe *what* you tried, what worked, what did not work, and *why* you think it did or did not work. For example, did you use an off-the-shelf method? How did you adapt it to the given task? Why did this adaptation work or not? Did you do some preprocessing or cleaning of the dataset? Was it useful? Why or why not?

It is not enough to report the evaluation scores of your methods. You need to explain the reasons why you think you see these scores. You need to show in the report that you have tried to understand why you got these specific evaluation scores, regardless of whether they are high or low.

You should also describe the limitations of what you did. What could have led to improvements in model performance? How could you have approached this task differently? What was particularly challenging about working with this dataset and why do you think that was? This discussion does not require further experiments, but requires you to examine your experimental results, critically think about your choices and the assumptions you made, hypothesise on how you can overcome some limitations and improve your solution. Your discussion should be based on evidence, such as lecture material and relevant literature.

Finally, in all cases, you should cite relevant literature which informed your choices in terms of modeling, analysis, etc.

Academic Code of Conduct

Discussion with fellow students regarding the project is allowed, but sharing of code or written content is strictly prohibited. Any instances of directly copying code or text from other students will be considered as plagiarism and will be subject to the University's plagiarism regulations.

Extra work

The above is a description of the minimum that we expect you to implement, but you are allowed to expand upon this to explore more complex ideas. If you decide to submit extra work, that extra work will be graded as well.

References

- [1] Z. Dai and J. Callan. Context-aware Sentence/Passage Term Importance Estimation for First Stage Retrieval. *arXiv preprint arXiv:1910.10687*, 2019. <https://arxiv.org/abs/1910.10687>.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018.
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics.
- [4] R. Jagerman, H. Zhuang, Z. Qin, X. Wang, and M. Bendersky. Query expansion by prompting large language models. *arXiv preprint arXiv:2305.03653*, 2023.
- [5] S. Kuzi, A. Shtok, and O. Kurland. Query Expansion Using Word Embeddings. In *Proceedings of the 25th ACM International on Conference on Information and Knowledge Management, CIKM '16*, page 1929–1932, New York, NY, USA, 2016. Association for Computing Machinery. http://publish.illinois.edu/saar-kuzi/files/2017/10/w2v_cikm16.pdf.
- [6] N. Reimers and I. Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 11 2019.
- [7] L. Wang, N. Yang, and F. Wei. Query2doc: Query expansion with large language models. *arXiv preprint arXiv:2303.07678*, 2023.